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OPERATING LAMP ASSEMBLY COMPRISING AN AUTOMATICALLY ORIENTABLE CAMERA

Abstract

The present disclosure relates to an operating lamp assembly which comprises at least one operating lamp for illuminating an operating field, having at least one camera for acquiring an image of the operating field. The operating lamp assembly comprises a controller which automatically orientates the image of the camera.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. Non-Provisional patent Ser. No. 17/661,874, entitled “OPERATING LAMP ASSEMBLY COMPRISING AN AUTOMATICALLY ORIENTABLE CAMERA”, filed on May 3, 2022. U.S. Non-Provisional patent Ser. No. 17/661,874 claims priority to German Patent Application No. 10 2021 111 459.4, filed May 4, 2021. The entire contents of each of the above-listed applications are hereby incorporated by reference for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to an operating lamp assembly, which comprises at least one operating lamp having at least one camera.

BACKGROUND

[0003] An operating theatre is frequently equipped with a camera orientated onto the operating field, wherein the image of the camera is transferred onto a monitor in the operating theatre or in an external room, for example for training purposes. As a result, the operating field can also be made visible for people who do not have a direct view into the operating field. It is likewise conceivable for the image of the camera to be of use to the surgeon when performing the operation. Furthermore, the image of the camera can be displayed for documentation purposes.

SUMMARY

[0004] It is conventional to provide the camera on an operating lamp, wherein the operating lamp serves to illuminate the operating field. However, when orientating the operating lamp initially no notice is taken of the orientation of the camera. This may lead to the undesired situation of the image of the camera not corresponding to the desired orientation, for example of a viewing direction of the surgeon, and the operating field being shown distorted. The image may be on its head for example. This makes it orientation on the image of the camera virtually impossible.

[0005] In order to solve this problem, the camera was or is hitherto orientated manually. A disadvantage in this case is that, in addition to the manual orientation of the operating lamp, additional manual orientation of the camera is required.

[0006] The object of the present disclosure is therefore that of providing an improved operating lamp assembly.

[0007] This object is achieved by an operating lamp assembly, and/or a set composed of an operating lamp assembly and a symbol, and/or a method for operating an operating lamp assembly.

[0008] The present disclosure comprises, in a first aspect, an operating lamp assembly, which comprises at least one operating lamp for illuminating an operating field, having at least one camera for acquiring an image of the operating field. A controller of the operating lamp assembly is designed such that an image of the camera is automatically orientated.

[0009] Automatic orientation of the image makes it possible to avoid the outlay required hitherto for manual orientation. It is furthermore possible for example to avoid the situation where the observer of the image cannot orientate himself towards the image on account of an incorrectly orientated camera. Furthermore, the problem arising in the case of manual orientation, that either a controller has to be provided in the sterile region, or the surgeon cannot orientate the camera himself, is overcome.

[0010] In a possible embodiment of the present disclosure, the camera is arranged on the operating lamp, in particular on a handhold of the operating lamp.

[0011] In a possible embodiment of the present disclosure, the camera is arranged on a carrier system of the operating lamp assembly.

[0012] In a possible embodiment of the present disclosure, the orientation of the optical main axis of the camera relative to the optical main axis of the operating lamp is not automatically adjustable. Optionally, however, an orientation of the optical main axis of the camera relative to the optical main axis of the operating lamp may be manually adjustable.

[0013] However, the present disclosure is used in operating lamp assemblies in which the orientation of the optical main axis of the camera relative to the optical main axis of the operating lamp is specified in a structurally fixed manner.

[0014] In a possible embodiment of the present disclosure, an optical main axis of the camera is parallel with or at a fixed angle with respect to the optical main axis of the operating lamp. In particular, the relative orientation of the operating lamp and the camera is selected such that the camera acquires an image of the region illuminated by the operating lamp.

[0015] In a possible embodiment of the present disclosure, the optical main axis of the camera coincides with the optical main axis of the operating lamp.

[0016] In particular, the camera can be arranged on a handhold that is arranged centrally on the operating lamp, and be orientated in the same direction as the operating lamp.

[0017] In a possible embodiment of the present disclosure, the orientation of the image of the camera is achieved by orienting the camera.

[0018] In a possible embodiment of the present disclosure, the orientation of the image of the camera is achieved by digital orientation of the image.

[0019] In particular, the orientation of the image of the camera can be achieved by means of rotating the camera about at least one axis.

[0020] In a possible embodiment of the present disclosure, the orientation of the image of the camera is achieved by means of rotating the camera about the optical main axis thereof.

[0021] In a possible embodiment of the present disclosure, the orientation of the image of the camera is achieved by means of digital rotation of the image, in particular by means of digital rotation of the image in the image plane or about an axis which is perpendicular on the image plane.

[0022] In a possible embodiment of the present disclosure, the operating lamp assembly comprises a drive which can be actuated by means of the controller such that it brings about an orientation of the camera, in particular a rotation of the camera about the optical main axis thereof, such that the image of the camera is orientated.

[0023] In a possible embodiment of the present disclosure, the drive is arranged on the operating lamp.

[0024] In a possible embodiment of the present disclosure, the drive is integrated in the operating lamp.

[0025] In particular, the drive can be arranged in the handhold of the operating lamp, and the camera can rotate about the optical main axis thereof, which coincides with the optical main axis of the operating lamp.

[0026] In a possible embodiment of the present disclosure, the controller determines an orientation of the camera by means of an inclination sensor.

[0027] In a possible embodiment of the present disclosure, the inclination sensor is provided on the operating lamp.

[0028] In a possible embodiment of the present disclosure, the inclination sensor is provided on a carrier system of the operating lamp assembly.

[0029] The inclination sensor ascertains the inclination with respect to the vertical direction specified by gravity, and therefore determines an absolute orientation of the element on which it is arranged, in space, in particular the absolute orientation of the operating lamp.

[0030] In a possible embodiment of the present disclosure, the controller determines an orientation

of the camera by means of an object detection means for detecting objects in the image of the camera.

[0031] In a possible embodiment of the present disclosure, the controller determines a target orientation, in at least one operating mode, on the basis of signals of an inclination sensor, and automatically orientates the image accordingly.

[0032] In a possible embodiment of the present disclosure, in the target orientation, the image of the camera is orientated such that an upper edge of the image of the camera is orientated so as to be in parallel with the horizontal.

[0033] In a possible embodiment of the present disclosure, the controller determines a target orientation, in at least one operating mode, by means of an object detection means, and automatically orientates the image accordingly.

[0034] In a possible embodiment of the present disclosure, the object detection means detects one or more objects in the image of the camera.

[0035] In particular, the object detection means may comprise image recognition software stored in memory. In particular, the object detection means may be image recognition software stored in memory and configured to process image data.

[0036] In a possible embodiment of the present disclosure, the object is a symbol, in particular an arrow, which defines a direction, wherein the controller automatically orientates the image according to said direction.

[0037] In a possible embodiment of the present disclosure, when the image of the camera is in the target orientation, the direction of the symbol corresponds to a direction from an upper edge to a lower edge of the image.

[0038] In a possible embodiment of the present disclosure, when the image of the camera is in the target orientation, the direction of the symbol corresponds to the vertical of the image.

[0039] In a possible embodiment of the present disclosure, the symbol is deposited on a patient, beside an operating field, and/or arranged on a medical instrument.

[0040] In a further possible embodiment of the present disclosure, the object is a person, in particular a surgeon.

[0041] In a possible embodiment of the present disclosure, the controller automatically orientates the image according to a viewing direction of the person and/or a gesture of the person.

[0042] In a further possible embodiment of the present disclosure, the object is a medical instrument and/or an assembly composed of a plurality of medical instruments, for example one or more retractors, wherein the controller orientates the image according to the orientation of the instrument and/or of the assembly of a plurality of medical instruments.

[0043] In a further possible embodiment of the present disclosure, the object detection means identifies an OP situation, wherein the controller orientates the image in accordance with the identified OP image.

[0044] For example, the people involved and their arrangement around the patient are identified, and the image is orientated according to a target orientation for the identified OP situation. A plurality of OP situations, together with respective target orientations of the image, may be stored in the controller. In a possible embodiment of the present disclosure, in at least one operating mode the image can be orientated manually, and/or in at least one operating mode the controller orientates the image on the basis of signals of a manually operable input device.

[0045] Furthermore, in at least one operating mode the image may be manually orientable, wherein the manual orientation takes place independently of the controller, in particular by manual rotation of the camera.

[0046] In a possible embodiment of the present disclosure, the operating lamp assembly has at least two operating modes, wherein in a first operating mode the orientation of the image takes place automatically, and in a second operating mode the orientation of the image takes place manually. In the first operating mode, the orientation can take place on the basis of signals of an inclination

sensor and/or by means of an object detection means.

[0047] In a possible embodiment, the controller is designed such that an operator can select among a plurality of different operating modes for automatic and/or manual orientation of the image.

[0048] In particular, the operating lamp assembly can be designed such that it is possible to select from at least two operating modes, by means of manual operation of an input device.

[0049] In a second aspect, the present disclosure comprises a set composed of an operating lamp assembly and a symbol which is detected by an object detection means of the controller and is consulted for orientating the image, wherein the symbol may constitute a sterilisable element.

[0050] In particular, the symbol may be a single-use product.

[0051] In a third independent aspect, the present disclosure comprises a method for operating an operating lamp assembly, wherein the operating lamp assembly comprises at least one operating lamp and at least one camera. The method comprises the following steps: [0052] illuminating an operating field by means of the operating lamp; [0053] recording an image of the operating field by means of the camera; and [0054] automatically orientating the image of the camera, in particular by rotating the image of the camera, in particular by automatic rotation about a main axis of the camera.

[0055] In a possible embodiment of the present disclosure, the automatic orientation is performed by means of an inclination sensor and/or an object detection means for automatically orientating, in particular rotating, the image of the camera by means of the controller.

[0056] The method may be performed as has already been described in view of the operating lamp assembly according to the disclosure.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0057] The present disclosure will now be explained in greater detail with reference to drawings and embodiments.

[0058] In the drawings:

[0059] FIG. 1 shows an embodiment of an operating lamp assembly according to the disclosure comprising a camera;

[0060] FIG. 2 shows the embodiment of an operating lamp assembly according to the disclosure shown in FIG. 1, wherein an image, which was acquired by the camera, is displayed in a distorted orientation;

[0061] FIG. 3 shows the embodiment from FIG. 2, wherein the image has been orientated according to a target orientation.

DETAILED DESCRIPTION

[0062] FIG. 1 to FIG. 3 each show an embodiment of an operating lamp assembly 1 according to the disclosure, comprising a first operating lamp 2 and a second operating lamp 2'. However, within the scope of the present disclosure, the operating lamp assembly 1 could also comprise just one operating lamp 2 or more than two operating lamps 2.

[0063] In the respective embodiments, a carrier system 3 is provided, on which the operating lamps 2 and 2' are arranged above an operating table 8. The position and orientation of the operating lamps 2 and 2' can be adjusted. An adjustment can take place manually. Furthermore, an adjustment can be achieved by means of one or more drives of the carrier system 3.

[0064] In the respective embodiments, the carrier system 3 comprises a ceiling mount (not shown), on which a central shaft (not shown) can be arranged. Carrier arms 5 can in turn be pivotably arranged on the central shaft, wherein the carrier arms may be pivotable about a vertically extending axis formed by the central shaft. The operating lamps 2 and 2' can be arranged on different carrier arms 5 in each case, by means of further carrier arm elements 6 and/or joints.

Embodiments of the carrier system **3** other than those just described are also conceivable.

[0065] The operating lamps **2** and **2'** shown in FIG. **1** to FIG. **3** each generate a light field. The operating lamps **2** and **2'** can now be orientated and/or positioned such that an operating field **F** of a patient **P**, lying on the operating table **8**, is illuminated as desired. In other cases, the light fields of the two operating lamps **2** and **2'** can also be orientated on different regions. This is conceivable for example in the case of a transplant, wherein an operating lamp **2** can be directed onto the operating field **F**, while an operating lamp **2'** can be directed onto the organ to be transplanted.

[0066] In the embodiment shown in FIGS. **1** to **3**, the operating lamps **2** and **2'** comprise a handhold **7** and **7'** for positioning and orientating the respective operating lamps **2** and **2'**. In FIG. **1**, only the handhold **7** of the operating lamp **2** is visible. In the embodiment of FIG. **1**, the handhold **7** is arranged centrally on the operating lamp **2**. The handhold **7** is in particular arranged centrally on a lamp body and surrounded by light-emitting regions of the lamp. The main axis of the handhold **7** and **7'**, respectively, may correspond to the optical main axis of the lamp **2** and **2'**, respectively.

[0067] The handhold **7** and **7'**, respectively, can furthermore, as shown in the embodiments shown in FIGS. **1** to **3**, be arranged in the geometric centre of a light source **2a'** of the corresponding operating lamp **2** and **2'**, respectively, that generates the light field. As shown in FIGS. **1** to **3**, the operating lamp **2** and **2'**, respectively, can also comprise one or more handhold elements, which are arranged laterally on the operating lamp **2** and **2''**, respectively. The handhold elements can be provided as an alternative to or in addition to the handhold **7** and **7'**.

[0068] In the embodiment shown in FIGS. **1** to **3**, the operating lamp assembly **1** comprises a camera **10** for generating an image **11**, wherein the operating lamp assembly **1** further comprises a monitor **60** for displaying the image **11** of the camera **10**. The camera **10** is arranged on the first operating lamp **2**.

[0069] In a first variant, the operating lamp **2** does not comprise a camera.

[0070] In contrast, in a second variant, the operating lamp assembly **1** comprises two cameras **10** und **10'**, which are arranged correspondingly on the operating lamps **2** and **2'**.

[0071] The monitor **60**, as well as the operating lamps **2** and **2'**, can be arranged on the carrier system **3** by means of a carrier arm element **6**. In the embodiments shown in FIGS. **1** to **3**, the monitor **60** is located in the operating theatre. It is also conceivable, however, for the monitor **60** or another monitor to be provided in a room that is separate from the operating theatre. In this way, the image **11** of the camera **10** can be made available for example for training purposes. It is also conceivable to supply the image of the camera to a memory unit, and record it there, for documentation purposes.

[0072] Within the meaning of the present disclosure, the term “image” includes both a video image within the meaning of an image sequence, and an individual image within the meaning of a photograph.

[0073] In a possible embodiment of the present disclosure, the image is a video image.

[0074] As shown in FIGS. **1** to **3**, in the embodiment the optical main axis **OH** of the camera **10** and the optical main axis **H** of the operating lamp **2** are orientated so as to be mutually parallel, and furthermore coincide. As shown in FIGS. **1** to **3**, the optical main axis **H** of the operating lamp **2** can extend through a geometric centre of the operating lamp **2**, in particular through a geometric centre of a light source of the operating lamp **2**, in particular through a centre point of the operating lamp **2**.

[0075] Viewed along the main axis thereof, the operating lamp **2**, in particular the light source thereof, can be rectangular and/or round in shape, and/or is formed in a plate-shaped manner.

[0076] As shown in FIGS. **1** to **3**, the optical main axis **H** of the operating lamp **2** corresponds to a central axis of a light cone generated by the light source. If both operating lamps **2**, **2'** each comprise a camera **10**, **10'**, the optical main axis of the relevant camera **10**, **10'** may coincide with the optical main axis of the relevant operating lamp **2''**.

[0077] According to the present disclosure, the operating lamp assembly comprises a controller

which is designed such that an image of the camera is automatically orientated.

[0078] In FIGS. 1 to 3, the relevant operating lamp assembly 1 comprises a controller 20, by means of which the image 11 of the camera 10, 10 is automatically orientated. This can be achieved by means of orientating the camera 10 or 10', and/or by means of digitally orientating the image 11 and 11', respectively, of the camera 10 and 10', respectively. The digital orientation of the image 11 or 11' can be achieved by means of the controller 20, in particular by means of an arithmetic unit 50, which may be integrated in the controller 20. The relevant operating lamp 2 can be linked to the controller 20 and/or the input device 40 in a wired and/or wireless manner.

[0079] The orientation of the image 11 may be achieved by means of orientating the camera 10, 10', i.e. b means of mechanical orientation of the camera 10, 10, wherein the mechanical orientation may be achieved by means of the drive 25, 25 which is shown in FIG. 1 for example. In the embodiment shown in FIGS. 1 to 3, the drive 25 is arranged on the operating lamp 2. In particular, the drive 25 can be integrated into the handhold 7 of the operating lamp 2, as shown in FIG. 1. By means of the controller 20, the drive 25, 25' can be actuated such that the camera 10, 10 is orientated according to a target orientation of the image 11, 11.

[0080] In the embodiment, the camera 10, 10 can be rotated about the optical main axis OH thereof, in order to orientate the image 11. For this purpose, the controller 20 actuates the drive 25, 25' such that a target orientation of the image 11, 11' of the camera 10, 10 is achieved by means of the orientation of the camera 10, 10.

[0081] Alternatively or in addition, the automatic orientation of the image 11 or 11' can be achieved by means of digitally orientating the image 11 or 11', in particular by rotating the image in the image plane.

[0082] The automatic orientation of the image can take place in different manners:

[0083] In a first embodiment, the controller 20 determines the orientation of the camera 10, 10' on the basis of signals of an inclination sensor 21. The inclination sensor 21 can be arranged on the operating lamp 2, 2', as shown for example in FIGS. 1 to 3. It is also conceivable, however, to provide the inclination sensor 21 at another location, for example on the carrier arm 5. The inclination sensor 21 may be a digital and/or electronic spirit level.

[0084] The image 11 or 11', respectively, may be orientated on the basis of the inclination sensor 21 or 21', respectively, such that an upper edge of the image 11 or 11, respectively, is orientated so as to be in parallel with a horizontal.

[0085] In a second embodiment, an object detection means 30 is provided, by means of which the controller 20 can determine the orientation of the image 11, 11' of the camera 10, 10', as is shown for example in FIGS. 1 to 3. The automatic orientation of the image 11 or 11' then takes place into a target orientation determined on the basis of the object detection means 30.

[0086] The object detection means 30 may be image recognition software stored in memory for example. The controller 20 may comprise the object detection means 30, in particular the object detection means 30 may be an integral component of the controller 20. However, the object detection means 30 may also be provided separately from the controller 20. The object detection means 30 can be actuated such that it can detect one or more objects in the image 11 of the camera 10 or in the image 11' of the camera 10.

[0087] The object detection means 30 can have one or more of the following functions: [0088] a. The OP situation is identified by means of the acquired image 11 or 11'', respectively, of the camera 10 or 10', respectively, and the image 11 or 11', respectively, is then orientated. For example, the people O and/or A involved and their arrangement around the patient P are identified, and the image 11 or 11' is orientated according to a target orientation for the identified OP situation. A plurality of OP situations, together with respective target orientations of the image 11 or 11', can be stored in the controller 20. [0089] b. The surgeon O is identified by means of the acquired image 11 or 11, respectively, and the image 11 or 11', respectively, is orientated according to the surgeon's viewing direction. It is in particular possible to identify on which side of the operating field F the

surgeon O is located, and the image **11** and/or **11'** is orientated such that the vertical of the image **11** and/or **11** corresponds to a connecting line between the position of the surgeon O and the operating field F. In this case, the direction of the connecting line between the position of the surgeon O and the operating field F can be defined as the viewing direction of the surgeon O. Alternatively, the actual viewing direction of the surgeon can be identified, and can form the basis of the automatic orientation. [0090] c. A medical instrument (e.g. retractor, clamp or endoscope) is identified by means of the acquired image **11** or **11'**, respectively, and the image **11** or **11'**, respectively, is orientated according to the direction and/or position of said instrument. [0091] d. An accessory/tool is identified by means of the acquired image **11** or **11'**, respectively, according to which accessory/tool the image **11** or **11**, respectively, is intended to be orientated. This may for example be a direction pointer, which may be designed so as to be sterilizable or as a single-use product. Alternatively, the orientation of a passive panel, which is in any case detected for actuating the light assembly **1** and/or other devices in the operating theatre, which is consulted for orientating the image **11** or **11'**, respectively, of the camera **10** or **10**, respectively.

[0092] The passive panel according to variant d) may in particular be a passive operating panel, as is known from DE 10 2020 114 426 by the applicant. The operating panel may be a passive, sterile and/or sterilizable element. At least one symbol may be provided on the operating panel, by means of which symbol operator inputs can be performed on the operating panel. The operating panel may be a completely passive operating panel, i.e. it does not have any electronic components or a power supply. It may be produced from a sterilizable plastics material or stainless steel. On account of these properties, it is sterilizable without problem, and is prepared in a sterilized manner just like the normal instruments such as scalpels, clamps, scissors, etc. On account of the very simple structure, a single-use variant is also conceivable.

[0093] In points a.-d., the evaluation takes place by means of image recognition. Resulting herefrom, the drive **25** or **25'**, respectively, of the camera **10** or **10**, respectively, is controlled, and the image **11** or **11'**, respectively, is automatically orientated. It is also possible to digitally orientate/rotate the image **11** or **11'**, respectively, and no drive **25** or **25'**, respectively, is required for this purpose. The rotation is then performed exclusively digitally.

[0094] As shown in FIG. **1**, according to variant d), in addition to the operating field F, a symbol S, which, as shown, may be an arrow, can be provided. In the embodiment, the object detection means **30** detects the symbol S in the image **11** of the camera **10**, and determines the orientation of the camera **10** and/or the orientation of the image **11** on the basis of the orientation R of the symbol S. The controller **20** then orientates the image **11** of the camera **10** in accordance with a target orientation, on the basis of the determined orientation R of the symbol S. The symbol S can for example also be provided on a medical instrument. For example, the image **11** can be orientated such that the symbol S points in a defined direction, for example upwards.

[0095] FIG. **2** shows an image **11** of the camera **10** in an orientation which does not correspond to its target orientation. In contrast, FIG. **3** shows an image **11** of the camera **10** which is orientated in accordance with a target orientation. The image **11** shown in FIG. **3** was obtained by means of orientation according to the disclosure, in particular rotation, of the image **11** shown in FIG. **2**. The rotation of the image **11** can be preformed by rotating the camera **10** or by digital rotation of the image of the camera.

[0096] In a second variant, the object detection means **30** detects the surgeon O on the image **11** of the camera **10**. In particular, the object detection means **30** can detect the viewing direction B of the surgeon O, wherein the image **11** of the camera **10** is orientated according to the viewing direction B, by means of the controller **20**. It is also conceivable for the image **10** of the camera **11** to be orientated according to a detected gesture of the surgeon O.

[0097] The images **11** and **11** shown in FIG. **2** and FIG. **3** in each case show an operating field F, wherein the viewing direction B of the surgeon O is illustrated by means of an arrow. Furthermore, a patient P is shown.

[0098] The optical main axis OH of the camera, by means of which the image **11** was acquired, extends in both figures perpendicularly with respect to the image plane of the display, and thus with respect to the drawing plane in the figures. As shown in FIGS. **2** and **3**, the optical main axis OH of the camera, about which the image is rotated, can extend through the geometric centre, in particular the centre point, of the image **11**.

[0099] The orientation/rotation of the image **11** can be preformed for example by rotating the camera **10** and/or by means of digital rotation of the image **11** of the camera **10**.

[0100] In the embodiment shown in FIG. **3**, the target orientation corresponds to an orientation of the image **11** in which the viewing direction B of the surgeon O detected by the image detection means **30** extends from a lower edge to an upper edge of the image **11**.

[0101] The operating lamp assembly may have a plurality of operating modes, in which the automatic orientation take place in each case on the basis of different criteria. The criterion to be used in each case can be selectable by an operator of the operating lamp assembly.

[0102] The embodiment of the operating lamp assembly **1** shown in FIGS. **1** to **3** has a first operating mode, in which the orientation of the image **11**, **11'** takes place automatically on the basis of signals of the inclination sensor **21**. In the embodiment, the operating lamp assembly **1** furthermore has a second operating mode, in which the orientation of the image **11** takes place automatically, by means of the object detection means **30**. In the embodiment, the operating lamp assembly **1** has a third operating mode, in which the orientation of the image **11** takes place manually.

[0103] In an embodiment of the operating lamp assembly **1** comprising two cameras **10** and **10'**, the orientation of the respective images **11**, **11'** can take place in a mutually independent manner, according to different operating modes or according to a common operating mode.

[0104] In the first operating mode of the operating lamp assembly **1** shown in FIGS. **1** to **3**, the controller **20** thereof determines the orientation of the camera **10** by means of the inclination sensor **21** which is arranged on the operating lamp **2**. On the basis of the signals of the inclination sensor **21**, the controller **20** determines a target orientation of the image **11** and automatically orientates the image **11** accordingly.

[0105] In the second operating mode of the operating lamp assembly **1** shown in FIGS. **1** to **3**, the orientation of the image **11** is determined by means of the object detection means **30**. A target orientation of the image **11** is determined by the object detection means **30**, wherein the image **11** is automatically orientated accordingly.

[0106] The second operating mode may comprise a plurality of sub-modes, which differ with respect to the criteria according to which an automatic orientation takes place on account of the object detection means **30**. In particular, depending on the sub-mode, the automatic orientation can take place on the basis of various criteria evaluated by the object detection means, in particular on the basis of the criteria a) to d) above.

[0107] Both in the first and in the second operating mode, the automatic orientation of the image **11** by means of the controller **20** can take place by actuating the drive **25**, and a mechanical orientation of the camera **10** which corresponds thereto.

[0108] In the embodiment, for this purpose the camera **10** is rotated about the optical main axis OH thereof, such that the orientation of the camera **10** corresponds to the relevant target orientation of the image **11** of the camera **10**.

[0109] Alternatively or in addition, the automatic orientation of the image **11** can be achieved by means of digital orientation, in particular rotation, of the image **11**. The digital orientation of the image **11** can be achieved by means of the controller **20**, in particular by means of an arithmetic unit **50** of the controller **20**. Alternatively, the digital orientation of the image **11** can also be achieved by means of an arithmetic unit **50** which may be provided separately from the controller **20**. The controller **20** can communicate with the arithmetic unit **50** wirelessly and/or in a wired manner.

[0110] The controller may include a processor and memory with instructions stored therein to carry out the operations described herein. The arithmetic unit may include a processor and/or application specific integrated circuit (ASIC) programmed to carry out arithmetic operations as described herein.

[0111] In the third operating mode of the operating lamp assembly **1** shown in FIGS. **1** to **3**, the orientation of the image **11** takes place manually. In the embodiment, this takes place by means of manual actuation of the input device **40**.

[0112] In the embodiment, the input device **40** communicates with the controller **20**, wherein the controller **20** orientates the image **11** on the basis of signals of the input device **40**. The manual orientation of the image **11** can take place by means of orientating the camera **10**, in particular rotating the camera **10** about the optical main axis OH thereof, and/or by means of digitally orientating the image **11**, in particular by means of rotating the image **11** about the optical main axis OH thereof.

[0113] It is also conceivable to design the operating lamp assembly **1** such that, in one operating mode, manual orientation of the image **11** takes place in combination with automatic orientation of the image **11**. In an operating mode of this kind, the controller **20** could tailor the two types of orientation to one another. It would be conceivable, for example, for manual orientation of the image **11** to be used only for readjustment, for example after automatic orientation of the image **11** has taken place.

[0114] An operator can select among the different operating modes of the operating lamp assembly.

[0115] An operator can select among different operating modes of the operating lamp assembly such that at least two images of different cameras are orientated in accordance with different operating modes or in accordance with a common operating mode.

[0116] In order to switch among the different operating modes, the input device **40** shown in FIGS. **1** to **3** may be provided, by means of the manual operation of which the different operating modes can be selected.

[0117] For this purpose, the input device **40** shown in FIGS. **1** to **3** may comprise at least one input element **41**. In the embodiments shown in FIGS. **1** to **3**, the input device **40** is furthermore designed such that the manual orientation of the image **11** takes place by means of the manual actuation of the input device **40**. For this purpose, corresponding input elements **41** may be provided on the input device **40**, which elements may be provided for example in the form of switches, adjusters, and/or a touch screen. For example, a first input element **41** for rotating the image **11** to the left, about the optical main axis thereof, and a second input element for rotating the image **11** to the right, about the optical main axis thereof, may be provided. In an example, the input element may be a button or knob.

[0118] If two cameras **10** and **10'** are provided, the relevant orientation of the images **11**, **11'** can take place as already described above.

[0119] In an embodiment of the present disclosure, the images **11**, **11'** of the two cameras **10** and **10** are automatically orientated such that they show the operating field F in the same orientation.

[0120] Furthermore, the operating lamp assembly **1** may be designed such that it is possible to manually select, by means of the input device **40**, what image **11**, **11'** or what images are displayed on the monitor **60**. For example, it is thus possible for at least one image, from the images **11** and **11'**, to be selected for display on the monitor **60**, by means of manual actuation of the input device **40**. The images can for example also be arranged side-by-side on the monitor **60**.

LIST OF REFERENCE SIGNS

[0121] **1** operating lamp assembly [0122] **2**, **2'** operating lamp [0123] **2a** light source of the operating lamp **2'** [0124] **3** carrier system [0125] **5** carrier arm [0126] **6** carrier arm element [0127] **7** handheld [0128] **8** operating table [0129] **10**, **10'** camera [0130] **11**, **11'** image of the camera [0131] **20** controller [0132] **21** inclination sensor [0133] **25** drive [0134] **30** object detection means [0135] **40** input device [0136] **41** input elements [0137] **50** arithmetic unit [0138] **60** monitor

[0139] F operating field [0140] OH optical main axis of the camera **10** [0141] H optical main axis of the operating lamp **2** [0142] S symbol [0143] R direction [0144] O surgeon [0145] B viewing direction [0146] A assistant [0147] P patient

Claims

1. A method for operating an operating lamp assembly, the operating lamp assembly comprising at least one operating lamp, at least one camera, a drive to bring about an orientation of the at least one camera, an inclination sensor and a controller, wherein the method comprises: illuminating an operating field by means of the at least one operating lamp; and acquiring an image of the operating field by means of the camera; the controller performing the following steps: determining, by means of the inclination sensor, a target orientation of the image; and activating the drive of the camera to orient the camera according to the target orientation of the image.
2. The method of claim 1, wherein the image is displayed on a monitor.
3. The method of claim 1, wherein said controller determines the target orientation of the image based on the absolute orientation of the at least one operating lamp or a carrier system carrying the at least one operating lamp, as determined by the inclination sensor, with respect to the vertical direction specified by gravity.
4. The method of claim 1, wherein the drive rotates the camera about at least one axis.
5. The method of claim 4, wherein the at least one axis is colinear to an optical main axis of the camera.
6. The method of claim 1, whereby, in the target orientation, the image of the camera is orientated such that an upper edge of the image of the camera is orientated so as to be in parallel with the horizontal.
7. A method for operating an operating lamp assembly, the operating lamp assembly comprising at least one operating lamp, at least one camera and a controller, wherein the method comprises: illuminating an operating field by means of the at least one operating lamp; and acquiring an image of the operating field by means of the camera; the controller performing the following steps: determining, by an object detection means configured to detect one or more objects in the image, the orientation of the camera; determining, on the basis of said object detection means, a target orientation for the image of the camera, and digitally orientating the image to match the target orientation.
8. The method according to claim 7, wherein the image is displayed on a monitor.
9. The method of claim 7, wherein the digital orientation of the image is performed by means of an arithmetic unit being part of the controller.
10. The method of claim 7, wherein the digital orientation of the image is achieved by rotating the image in the image plane or about an axis which is perpendicular to the image plane.
11. The method of claim 7, wherein, in the target orientation, the image of the camera is orientated such that an upper edge of the image of the camera is orientated so as to be in parallel with the horizontal.
12. The method of claim 7, wherein the one or more objects are exactly one object, the exactly one object being a symbol defining a direction, wherein the controller automatically orientates the image according to said direction.
13. The method of claim 12, wherein, when the image of the camera is in the target orientation, the direction of the symbol corresponds to a direction from an upper edge to a lower edge of the image.
14. The method of claim 13, wherein, when the image of the camera is in the target orientation, the direction of the symbol corresponds to the vertical of the image.
15. The method of claim 13, wherein the symbol is deposited on a patient, beside an operating field, or arranged on a medical instrument.
16. The method of claim 7, wherein the object is a person, whereas the target orientation is

determined based on the viewing direction and/or the gesture of the person.

17. The method of claim 7, wherein the object is at least one medical instrument, whereas the target orientation of the image is determined based on the orientation and/or the assembly of the at least one medical instrument.

18. The method of claim 7, wherein the object is an OP situation, whereas the target orientation of the image is determined based on people involved and their arrangement around a patient.
